DE RUEDA CONSTRUCTION

Detailed Website & Construction ERP Specification

Purpose
Detailed requirements for the public company website, private construction ERP, roles, HR, attendance, payroll, PayMongo payment integration, projects, materials, finance, reports, security, UI, and the reusable Secret Recipe.

Brand standard

Use the preferred glossy green de RUEDA architectural emblem and wordmark with the CONSTRUCTION subtitle. Palette: green, dark blue, light blue, white, and black, with pink glitter only as a restrained accent. Existing mockups guide layout but do not prove live implementation.

Payment provider decision: PayMongo. This document intentionally excludes references to the previously considered providers and wallet-specific hubs. Confirm exact PayMongo capabilities against the company account and current official documentation before promising a feature.

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Truthfulness rule: Distinguish design mockup, implemented feature, test-mode integration, and verified production feature. Never present sample data or simulated transaction status as live.

1. Product vision and implementation rules

Build a unified browser-based platform with a public corporate website and a private, authenticated construction operations system. All sites share the same core features, while access is restricted by role and assigned site.

Core requirements

- Centralize HR, attendance, weekly payroll, deductions, projects, materials, expenses, documents, messages, public intake, and reports.
- Use a unique Employee ID as the primary employee key; names may repeat but IDs may not.
- Preserve history. Archive/inactivate rather than destructively delete employees, projects, and financial records.
- Use shared records so approved attendance feeds payroll and reports without duplicate data entry.
- Every feature must show real persisted data or be clearly labeled demo/placeholder.

2. Public company website

The public website represents the company and must not expose private ERP data.

Pages

- Home hero with approved logo, company introduction, and clear calls to action.
- About Us with approved company profile and values.
- Services with construction capabilities and descriptions.
- Projects/portfolio containing only approved public project summaries and photos.
- Hiring form for applicant contact details, role interest, and permitted attachments.
- Business Inquiry form for customers, suppliers, subcontractors, and partnerships.
- Contact page with official public contact details.
- System Login entry point to the private ERP.

Form behavior

- Validate fields and show understandable success/error messages.
- Save applications/inquiries to the approved database or configured company Drive workflow; restrict review to authorized staff.
- Use spam protection, rate limits, file type/size checks, and appropriate privacy notices.
- Track internal statuses such as New, Under Review, Contacted, Shortlisted, Closed, and Archived.
- Public visitors must never be able to browse submitted applications or private records.

3. Authentication, roles, permissions, and CEO controls

All authorization must be enforced on the server, not just by hiding UI controls.

Account security

- Support authorized account creation/invitations, login, logout, recovery, session expiry, and revocation.
- Use secure password hashing; never store plaintext passwords or hard-code a default CEO password.
- Offer multi-factor authentication and passkeys/biometrics where supported, with secure fallback/recovery.
- CEO first-time activation must verify identity, establish secure credentials and recovery, and create an audit record.
- Log failed sign-ins, sign-ins, sign-outs, recovery, role changes, and privileged actions.

Roles

- CEO/Owner: company-wide oversight, final approvals, delegation, role administration, and audit access.
- Admin 1/HR: employee records, onboarding, attendance review, payroll preparation, and permitted HR workflows.
- Admin 2/Operations or Payroll: assigned operational functions and sites only.

- Site Supervisor/Site Head: assigned-site attendance, daily reports, material receipt/usage, progress, and exceptions.
- Finance/Approver: approved expense/payment/reconciliation functions.
- Read-only/Auditor: view permitted records and reports without edit or approval rights unless explicitly granted.

Delegation and permissions

- Scope permissions by role, site, data category, and action (view/create/edit/approve/export/archive).
- CEO delegations must identify delegate, exact scope, effective dates, and revocation; do not silently grant all CEO rights.
- Restrict SSS/PhilHealth IDs, payroll, documents, payment settings, and sensitive exports.
- Every privileged API action must independently check authorization.

4. Executive dashboard

Provide consolidated company view and site-filtered view, with each metric linked to its underlying records.

Dashboard widgets

- Employee headcount, new hires, inactive employees, and distribution by site.
- Today's attendance: present, absent, late, missing time-out, and pending corrections.
- Weekly payroll states: draft, review, approved, and verified payment status where implemented.
- Project totals by ongoing, paused, completed, cancelled/unfinished, and archived.
- Materials alerts: low stock, reserved quantities, pending deliveries, and site requests.
- Expenses, budgets, pending approvals, and transactions requiring proof/reconciliation.
- Customer progress reports due, pending review, sent history, and delivery failures where available.
- Recent audit activity and unresolved operational exceptions.

Controls

- Date range and site/project filters; department filter where relevant.
- Cards must open the relevant filtered module, not act as decoration.
- Show last-updated timestamps and distinguish live from cached values.
- Do not show balances or payment results unless the backend has verified data.

5. Employee management and HR

Employee records must be searchable, unique, protected, and linked to historical attendance and payroll.

Profile fields

- Unique Employee ID; legal full name; job title/position; employment type; start date; employment status; site assignment; supervisor; daily rate or compensation basis.
- Contact details and emergency contact where company policy requires them.
- Employment documents, contracts, COE history, onboarding checklist, notes, and effective-dated changes.
- Protected SSS/PhilHealth and other statutory details visible only to authorized personnel.
- Payment destination data only when required by the approved workflow; mask it in ordinary screens.

Onboarding and edits

- Search for duplicate identities and assign a unique Employee ID.
- Validate required profile, rate, contact, and site fields; upload only permitted documents.
- Save, reopen, and verify the record; require approval for defined exceptions.
- Important edits must retain old/new values, editor, timestamp, and reason.
- Site transfers must have an effective date and preserve historical assignments.

Archive and rehire

- Termination changes status but does not erase attendance, payroll, deductions, or documents.
- Archive to an Archive/Junk Employee Records area while retaining authorized historical reports.
- Reactivation/rehire must link prior history and avoid duplicate identities.
- Deletion requires elevated authorization, retention review, and audit record.
- Generate COE using the approved template and company CEO name/title; preview before download or print.

6. Attendance, DTR, and site supervision

Attendance must preserve exact time records and corrections rather than silently overwriting them.

Capture

- Select date and site, identify employee by unique ID, and confirm the displayed profile.
- Support QR capture if implemented and authorized manual fallback. A scan is valid only if identity, timestamp, and site are saved server-side.
- Record exact clock-in/out timestamps to the minute; do not silently round or alter them.
- Apply company-configured break, payable-time, overtime, and exception rules.
- Record absence, late arrival, missed punch, overtime, and corrections with appropriate reason.

Review

- Supervisor reviews assigned-site entries and missing punches, duplicates, and unresolved exceptions.
- HR reviews exceptions before payroll cutoff; corrections require a reason and preserve original values.
- Search by date, site, employee ID, name, status, and exception type.
- Restrict edits to authorized site scope; log actor, time, and reason.

Outputs

- Daily/weekly DTR with employee, site, date, clock-in/out, break handling, hours, exceptions, reviewer, and correction history.
- Print-ready PDF and spreadsheet exports with selected filters and generated timestamp.
- End-of-day supervisor checklist for missing time-outs, absences, site conditions, progress, delays, and workforce concerns.

7. Weekly payroll, deductions, and payslips

Payroll calculations are drafts until verified and approved.

Calculation

- Select pay week/period and site or employee group.
- Use approved attendance and effective-dated compensation rates.
- Calculate base earnings and configured allowances/overtime/adjustments according to approved policy.
- Itemize SSS, PhilHealth, loans, cash advances, canteen balances, and other authorized deductions.
- Show gross, each deduction, total deductions, and net pay per employee and batch.
- Flag missing attendance/rates, duplicate entries, unusual changes, negative net pay, and inconsistent totals.

Review and finalization

- Create a draft payroll register and review exceptions before approval.
- Record approver, timestamp, period, batch total, and notes.
- Lock or version the approved payroll snapshot so later attendance edits do not silently change it.
- Use a controlled adjustment/reversal workflow for post-approval corrections.
- A payment request is not proof of completed payment; reconcile with verified provider status or documented manual evidence.

Payslips

- Generate individual/site/company payslips with pay period, earnings, itemized deductions, net pay, and verified payment reference/status when available.
- Protect individual payslips from access by other employees.
- Send notifications only after the configured trigger is confirmed; minimize sensitive data in SMS.
- Log notification destination, time, delivery status where available, and retry results.

8. PayMongo payments and transaction ledger

PayMongo is the selected payment provider. Integrate only products and methods enabled for the company account. Do not assume that incoming payment collection automatically includes employee payouts; verify the required use case against PayMongo's current official documentation and account permissions.

Scope

- Use PayMongo for supported incoming customer payments and payment status tracking.
- Use it for outgoing disbursements only if the required product/API and account explicitly support that use case.
- If a required payout method is not supported, provide a clearly labeled authorized manual process instead of faking an integration.
- Use a generic internal module name such as Payments & Transactions; no provider-specific wallet hub.

Payment lifecycle

- Create an internal record with unique ID, customer, invoice/project, amount, currency, purpose, and state.
- Create payment sessions/links on the backend using server-side credentials; expose only safe checkout data to the frontend.
- Store provider identifiers and expected amount/currency; reject mismatches and expired sessions.
- Receive webhook events over HTTPS and verify authenticity using PayMongo's current documented method.
- Process duplicate webhooks idempotently so they cannot create duplicate ledger entries or notifications.
- Use server-side status checks when events are delayed or ambiguous.
- Only mark paid after verified confirmation; distinguish pending, failed, expired, cancelled, refunded, and needs-review states where supported.
- Link confirmed payments to the correct customer/invoice/project and finance ledger; reconcile totals and discrepancies.

Security and ledger

- Store secret keys in server-side environment variables or a secrets manager; never expose them in frontend code or Git history.
- Separate test and live credentials and test before production.
- Restrict who may change payment settings, issue refunds where supported, or record manual payments.
- Ledger fields: internal ID, provider reference, customer/project/invoice, amount/currency, method, timestamps, status, reconciliation status, actor/approver, and notes.
- Keep payment collection and payroll as separate linked processes; receiving customer payment does not mean payroll has been paid.

9. Sites, projects, and customer progress reports

Every site has the same core structure; users see the sites authorized by their role.

Project record

- Project ID/name, customer/contact details, site address, manager/supervisor, budget/contract value, start and planned completion dates, status, and documents.
- Statuses: Draft, Ongoing, Paused, Completed, Cancelled/Unfinished, Archived, Restored.
- Record status changes with date, reason, actor, and notes; preserve completed/cancelled history.
- Link employees, contractors, materials, expenses, progress updates, issues, and attachments.
- Show schedule risks, budget use, blockers, and outstanding approvals where data exists.

Customer progress reporting

- Collect customer contact and preferred delivery method at project setup; restrict access to contact data.
- Generate milestone reports at 25%, 50%, 75%, and 100%, plus monthly reports between milestones.
- Include dated site photos, completed work, current activities, next steps, delays/issues, and prepared-by details.
- Generate customer-ready PDF; preview and manually approve before sending.
- Retain report version, period, photos, approver, recipient, sent time, delivery status, and resend attempts.
- Exclude internal costs, payroll, private notes, and unrelated employee information.

Site daily report

- Capture site conditions, workforce count, completed work, materials received/used, contractor/equipment issues, safety observations, blockers, photos, and follow-ups.
- Support draft, submitted, reviewed, returned, and approved states where configured.
- Require date and site to prevent orphaned records.

10. Materials, suppliers, contractors, and inventory

Track materials from request and purchase through delivery, site release, use, and adjustment.

Catalog and stock

- Material ID/name, specification, unit, category, default supplier, unit cost, reorder threshold, and active/archive status.
- Track on-hand, reserved, available, and in-transit quantities separately where implemented.
- Apply a documented and consistent stock valuation method.
- Low-stock alerts must link to the material and site.

Purchases and deliveries

- Requests include site, material, quantity, reason, needed-by date, supplier, and approval state.
- Track order reference, expected delivery, received/rejected/damaged quantities, cost, invoice/receipt, and receiver.
- Support partial delivery; do not mark a whole order received when only part arrived.
- Maintain supplier/contractor contacts, scope, related project, documents, delivery history, and notes.

Release and usage

- Record quantity issued, site, receiver, date/time, and reference.
- Record consumption, returns, and adjustments with reason and author.
- Prevent negative stock unless a controlled, auditable override is explicitly approved.
- Preserve movement history; do not silently rewrite past inventory movements.

11. Expenses, cash flow, documents, messages, and public intake

Supporting modules must link to projects/sites while enforcing privacy.

Expenses and cash flow

- Capture vendor, category, amount, date, site/project, purpose, receipt, requester, approver, and notes.
- Show budget versus actual, outstanding commitments, payroll liabilities, and categorized totals where supported by data.
- Require approval for configured thresholds and preserve approval history.
- Distinguish draft, pending, approved, confirmed paid, failed, refunded/reversed where applicable, and reconciled states.
- A receipt is supporting evidence, not by itself proof that a payment completed.

Documents and Drive

- Searchable document index with title, category, linked employee/project/site, owner, access classification, upload date, and file/link location.
- Google Drive may be used within the configured quota and sharing policy; do not claim control over a file unless implemented.
- Restrict links for employee, payroll, contract, and finance documents; avoid public sharing for sensitive files.
- Keep version/updated-date metadata where available.

Messages and public intake

- Internal direct/group messages with timestamps, site/project context, attachments, and read state if implemented.
- Keep sensitive HR/payroll files out of general operational groups.
- Route applications to HR and business inquiries to assigned staff; track status, reviewer, notes, follow-up, and archive.
- Validate public requests server-side and prevent them from accessing private ERP records.

12. Reports and audit trail

Reports must use saved data and clearly show filters and generation time.

Report catalog

- Executive summary; HR roster/onboarding/transfers/archive; DTR/absence/late/overtime/corrections; payroll register/deductions/net pay/payslips/reconciliation.
- Project status/progress/customer reports/schedule/budget; materials stock/low-stock/purchases/deliveries/releases/usage/suppliers.
- Finance income/payment records, expenses, outstanding items, refunds/disputes where supported, and reconciliation exceptions.
- Custom reports only where the field/filter builder can be secured and explained.

Filters and output

- Filter by date range, site/project, employee, status, supplier/customer, and category as applicable.
- Preview when practical; include title, period, applied filters, generated timestamp, and page numbers.
- PDF should be print-ready (letter size where required); Excel (.xlsx) should have useful headings and currency/date formats.
- Sensitive HR/payroll fields are excluded by default; require explicit permission to export protected data.
- Empty results must remain empty—never fabricate rows or chart values.

Audit activity

- Log actor, action, entity type/ID, timestamp, outcome, and reason/context for sensitive actions.
- Include security events, role changes, employee edits, attendance corrections, payroll approvals, payment changes, expense approvals, archive/restore, and exports.
- Restrict audit-log modification and access; do not log secret credentials.

13. Data model, validation, and cross-module rules

Use stable identifiers and explicit relationships so updates remain traceable.

Core entities

- User, Role, Permission, Delegation, Session/SecurityEvent.
- Employee, EmploymentHistory, EmployeeDocument, SiteAssignment, CompensationRate.
- Site, Project, Customer, Contractor/Supplier, Milestone, SiteReport, ProgressReport.
- AttendanceEvent, AttendanceException, AttendanceApproval, PayrollRun, PayrollLine, Deduction, Payslip, PayrollAdjustment.
- Material, InventoryMovement, PurchaseRequest, PurchaseOrder, Delivery, SiteRelease/Usage.
- Expense, Budget, PaymentRecord, ProviderEvent, ReconciliationRecord, refund/dispute records where supported.
- DocumentReference, Message, PublicApplication, BusinessInquiry, NotificationLog, AuditEvent.

Integrity rules

- Employee IDs are unique and stable; names are not keys.
- Attendance must reference a valid employee/date and the correct site when site-scoped.
- Payroll lines reference a payroll run, employee, compensation basis, attendance source, earnings, and deductions.
- Payment records have unique internal IDs and provider references when returned; state changes are traceable.
- Stock balances must be derivable from movement history and approved adjustments.
- Archived entities remain in authorized historical reports.
- Validate required fields, numeric ranges, allowed state transitions, and cross-record references on the server.

Time and money

- Store timestamps consistently and display Philippine local time to users where appropriate.
- Keep exact attendance times; configure payroll rules instead of hard-coding assumptions.
- Format Philippine peso amounts as ■ where applicable while retaining numeric precision in storage.
- Use a consistent timezone policy for attendance, reports, payment events, and audit logs.

14. Security, privacy, resilience, and recovery

Security

- Enforce server-side authorization for every private API, export, file, and privileged action.
- Use HTTPS, secure sessions/cookies, CSRF protection where applicable, input validation, rate limiting, and safe error messages.
- Mask sensitive identifiers and restrict government IDs, payroll, loans, cash advances, and personal documents.
- Store secrets server-side and rotate exposed credentials; never commit secrets to source control.

Resilience

- Back up the database and document metadata; define and test restore procedures.
- Handle network timeouts and webhook retries safely; a timeout is not proof of success or failure.
- Use duplicate protection for financial actions and external event processing.
- Keep operational logs useful without exposing private data.
- When provider services are unavailable, use pending/review states rather than inventing outcomes.

Privacy and retention

- Collect only necessary applicant/customer information and explain its use.
- Define retention/access rules for applications, former employees, payroll, receipts, contracts, and audit events.

- Keep internal cost/payroll information out of customer-facing reports.
- Avoid unrestricted public Drive links for sensitive records.

15. UI/UX and responsive behavior

Use the supplied mockups as the primary layout reference. Apply only agreed brand/color treatment unless a functional requirement requires a small change.

Visual standard

- Use the glossy green de RUEDA architectural emblem and CONSTRUCTION subtitle.
- Palette: green, dark blue, light blue, white, black; pink glitter remains a restrained accent.
- Maintain consistent sidebar, top bar, page titles, forms, tables, filters, cards, and status labels.
- Keep dense payroll tables legible; use adequate contrast, spacing, and clear typography.

Interaction standard

- Every visible control must work as labeled or be clearly marked unavailable.
- Provide loading, empty, success, error, validation, and permission-denied states.
- Confirm destructive/high-impact actions and support archive/restore where history matters.
- Confirm payroll approval, payment initiation, supported refund actions, and role changes.
- Prevent double-click duplicates and show reference numbers after accepted operations.
- Support keyboard navigation and labels; never rely on color alone to communicate status.

Mobile

- Support phone/tablet workflows for attendance, site reports, employee lookup, approvals, and documents.
- Stack cards, collapse navigation, and offer row-detail views on narrow screens.
- Show upload progress/retry for photos and attachments.
- The system remains browser-based unless a separate app is explicitly requested.

16. The Secret Recipe: npm, build, and deployment standard

This is the standing engineering rule. Editing code or seeing a screen render does not prove that the project is deployment-ready.

A. Project setup

- Standardize on npm. Do not use Bun as the project package manager.
- Remove conflicting Bun lockfiles (bun.lock or bun.lockb) when standardizing a repository on npm.
- Keep package.json and package-lock.json synchronized and commit the correct lockfile.
- Prefer npm ci in clean CI environments and keep changes focused.

B. Build gate

- Run npm run build after changes and fix all build errors before declaring the project ready.
- Do not claim the build passed without checking the actual command result.
- Confirm the real output directory and configure hosting to use it. For the established Cloudflare Pages setup, use dist/ when that is the configured output.
- Run configured lint/tests and check important routes, forms, assets, and runtime errors.

C. Deployment gate

- Verify repository, branch, commit, build command, output directory, and deployment logs.
- After deployment, test the published HTTPS URL on desktop and mobile.
- Check routes/deep links, refresh behavior, assets, form validation, authentication boundaries, and core workflows.
- Confirm the live URL reflects the latest intended build, not a stale cache or different branch.
- Record unfinished features, known issues, manual workarounds, and unconfigured integrations.

D. No fake functionality

- Never present mock data, UI-only controls, simulated payment status, or placeholder credentials as live.
- Use test mode until authorized production credentials and end-to-end verification are complete.
- A screen existing does not mean its backend, persistence, permissions, and error cases are implemented.
- Label each feature accurately: mockup, implemented/unconnected, test-mode integration, or verified production.

E. PayMongo rule

- Use current official PayMongo documentation for enabled products, API methods, webhook verification, and supported payment lifecycle.
- Keep secret keys server-side; verify webhook authenticity and process events idempotently.
- Confirm final status before ledger posting or payment-triggered notification.
- Test success, failure, cancellation/expiry, delayed and duplicate events, timeout, and reconciliation before production.

17. Acceptance tests and release checklist

Record each acceptance test as pass, fail, blocked, or not applicable. Untested requirements must not be marked passed.

Core acceptance scenarios

- Public pages work while private records remain protected.
- Authorized login/recovery works and unauthorized users cannot access protected APIs.
- HR can create a unique employee, edit permitted fields, transfer sites, archive, and restore while preserving history.
- Attendance stores exact timestamps; corrections preserve originals and record reason/actor.
- Payroll totals match approved attendance/rates/deductions; drafts cannot be mistaken for approved payroll.
- Payslips are restricted to authorized users and notifications follow verified triggers.
- PayMongo test flow creates payment records, verifies events, handles duplicates, and updates the ledger only after confirmation.

- Project/site reports, photos, milestones, materials, and expenses link to the correct project/site.
- Inventory matches movement history; partial delivery is accurately represented.
- Reports respect filters, export only permitted fields, and PDF/Excel totals match screen totals.
- Sensitive operations are auditable; the deployed site is tested on desktop and mobile.

Final release gate

- npm package files synchronized; no conflicting Bun lockfile.
- npm run build passes and configured tests/lint have been reviewed.
- Secrets are configured securely and absent from frontend bundles/repository.
- PayMongo test credentials, webhook verification, idempotency, and exceptions tested before live use.
- Backup and restore process documented; permissions/audit trail reviewed.
- Published HTTPS URL verified on desktop/mobile; key workflows work.
- Known limitations and mockup-only areas disclosed; no unsupported provider capabilities promised.

Definition of done

The relevant build succeeds; the deployed version is verified; sensitive actions are protected and auditable; data persists correctly; payment behavior matches the approved PayMongo capabilities; and no mockup or test result is misrepresented as production functionality.

> **Final implementation rule**
> Use the agreed UI mockups as the design reference, apply the approved de RUEDA brand, implement these workflows, use PayMongo for the approved payment integration, follow the npm-only Secret Recipe, and verify every release before declaring it ready.

This is a requirements specification, not proof that modules are implemented, PayMongo credentials are active, or deployment tests have passed.